

JOMAR L. RANGASAJO (BSChE-1B) PORTFOLIO 4: ITERATION (Interactive Smartwatch Activity Tracker)

In modern fitness technology, tracking physical activity is essential for monitoring health and lifestyle habits. Because walking paces and stride lengths vary from person to person, an interactive tracking system is needed to customize calculations. This program utilizes the input function paired to collect the user's specific walking data. The program then uses Iteration through a for loop to simulate a real-time tracking clock based on the inputted duration.

```
In [10]: walk_duration = int(input("Enter total walking time (in minutes): "))
steps_per_minute = int(input("Enter your walking pace (steps per minute): "))
stride_length = float(input("Enter your average stride length (in meters): "))
total_steps = 0
total_distance = 0.0
for minute in range(1, walk_duration + 1):
    total_steps = total_steps + steps_per_minute
    distance_this_minute = steps_per_minute * stride_length
    total_distance = total_distance + distance_this_minute
    print("Minute", minute, ": Steps = ", total_steps, ": Distance = ", total_distance)
print("Workout Summary")
print("Total Steps Taken:", total_steps, "steps")
print("Total Distance Traveled:", total_distance, "meters")
```

```
Minute 1 : Steps = 100 : Distance = 70.0 meters
Minute 2 : Steps = 200 : Distance = 140.0 meters
Minute 3 : Steps = 300 : Distance = 210.0 meters
Minute 4 : Steps = 400 : Distance = 280.0 meters
Minute 5 : Steps = 500 : Distance = 350.0 meters
Minute 6 : Steps = 600 : Distance = 420.0 meters
Minute 7 : Steps = 700 : Distance = 490.0 meters
Minute 8 : Steps = 800 : Distance = 560.0 meters
Minute 9 : Steps = 900 : Distance = 630.0 meters
Minute 10 : Steps = 1000 : Distance = 700.0 meters
Minute 11 : Steps = 1100 : Distance = 770.0 meters
Minute 12 : Steps = 1200 : Distance = 840.0 meters
Minute 13 : Steps = 1300 : Distance = 910.0 meters
Minute 14 : Steps = 1400 : Distance = 980.0 meters
Minute 15 : Steps = 1500 : Distance = 1050.0 meters
Workout Summary
Total Steps Taken: 1500 steps
Total Distance Traveled: 1050.0 meters
```

In []: